

Problem 4

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Problem 1 Two polynomials, h and g , are given

$$h(x) = \frac{x^2 - 4}{4}$$

$$g(x) = x - 2$$

State the form of the limit, evaluate the limit, or state that it does not exist. Justify your answer.

(a) $\lim_{x \rightarrow 2} \frac{h(x)}{g(x)}$

Explanation. $\lim_{x \rightarrow 2} \frac{h(x)}{g(x)} = \lim_{x \rightarrow 2} \frac{\frac{x^2 - 4}{4}}{x - 2}$

Form: $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{\frac{x^2 - 4}{4}}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{4(x - 2)} \\ &= \lim_{x \rightarrow 2} \frac{x + 2}{4} \\ &= \frac{4}{4} \\ &= 1 \end{aligned}$$

(b) $\lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3}$

Explanation. $\lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3} = \lim_{x \rightarrow 3} \frac{x - 2 - 1}{x - 3}$

Form: $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3} &= \lim_{x \rightarrow 3} \frac{x - 2 - 1}{x - 3} \\ &= \lim_{x \rightarrow 3} 1 \\ &= 1 \end{aligned}$$

$$(c) \lim_{x \rightarrow 4} \frac{h(x) - h(4)}{x - 4}$$

Explanation. $\lim_{x \rightarrow 4} \frac{h(x) - h(4)}{x - 4} = \lim_{x \rightarrow 4} \frac{\frac{x^2 - 4}{4} - 3}{x - 4}$

Form: $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 4} \frac{h(x) - h(4)}{x - 4} &= \lim_{x \rightarrow 4} \frac{\frac{x^2 - 4}{4} - 3}{x - 4} \\ &= \lim_{x \rightarrow 4} \frac{x^2 - 4 - 12}{4(x - 4)} \\ &= \lim_{x \rightarrow 4} \frac{x^2 - 16}{4(x - 4)} \\ &= \lim_{x \rightarrow 4} \frac{(x - 4)(x + 4)}{4(x - 4)} \\ &= \lim_{x \rightarrow 4} \frac{x + 4}{4} \\ &= 2 \end{aligned}$$